

The Memory Boom Behind AI

SK Hynix has become one of the clearest winners of the AI buildout. The South Korean memory-chip maker, once viewed as a deeply cyclical commodity business, is now benefiting from intense demand for high-bandwidth memory, the specialized DRAM used alongside advanced AI processors.

The change is dramatic. SK Hynix has historically operated in one of the most volatile parts of the semiconductor market, where oversupply can quickly turn profits into losses. The company was rescued by creditors in 2001, struggled for years to find a buyer, and was eventually acquired by SK Group in 2012 at a distressed valuation. As recently as 2023, it reported a large annual loss.

Today, the picture looks very different. Demand for AI infrastructure has made memory a strategic resource. Large language models require constant movement of vast amounts of data between memory chips and compute chips, especially GPUs. High-bandwidth memory helps reduce that bottleneck by stacking DRAM close to the processor, allowing data to move faster and more efficiently.

SK Hynix entered this market early. It began developing high-bandwidth memory years before the current AI boom, even though the investment once looked uncertain. What was previously seen as a difficult and possibly outdated product has become essential to modern AI accelerators.

That shift has transformed expectations for the company. SK Hynix is now expected to generate enormous profits, pay unusually large employee bonuses, and contribute significantly to South Korea's tax revenues. Its employees have also become cultural symbols of the country's new semiconductor wealth, with the company's status rising far beyond the technology sector.

Investors have responded accordingly. SK Hynix shares have risen sharply since 2019, while other memory producers have also benefited from the market's reassessment of DRAM and HBM suppliers. Micron, the only major U.S.-based DRAM producer, has also become central to the AI memory trade.

Still, the opportunity carries risk. DRAM remains cyclical, and memory buyers are already looking for ways to reduce usage. China is expanding production capacity, and new supply could eventually pressure prices. Even high-bandwidth memory, despite its technical complexity, is still tied to the broader economics of memory manufacturing.

For now, investors appear to believe the favorable cycle can continue for at least a few more years. AI infrastructure spending remains intense, U.S. hyperscalers are committing enormous capital to data centers, and agentic AI systems may require even more memory than earlier chatbot-style applications.

The central question is how long the shortage lasts. If AI demand continues to grow faster than memory supply, companies like SK Hynix and Micron may keep enjoying exceptional profits. If supply catches up or customers find ways to use less memory, the current boom could fade quickly.

For AI investors, memory has become impossible to ignore. The companies building the infrastructure may still dominate the headlines, but the suppliers of the memory behind that infrastructure are now just as important to watch.

Key Points

- SK Hynix has moved from a historically cyclical memory business into a central role in the AI supply chain.
- High-bandwidth memory is critical because AI workloads depend on fast movement of data between memory and processors.
- The company's early investment in HBM has become a major advantage.
- Surging profits may bring large employee bonuses and major tax revenue for South Korea.
- Micron and other memory producers are also benefiting from the AI memory cycle.
- The main risk is that DRAM remains cyclical, with future supply growth and efficiency improvements likely to pressure prices eventually.
- The AI boom may last longer if demand for memory keeps growing faster than production capacity.